

AIR – BORNE DISEASES

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Objectives

- At the end of this section, students will be able to:-
 - ❖ List common air born diseases.
 - ❖ Identify the common modes of air born diseases transmission.
 - ❖ Participate in diagnose and treatment of common air born diseases.
 - ❖ Apply preventive and control methods for air born diseases.

Introduction

- The organisms causing the disease in the air born group enter the body via the respiratory tract.
- During talking, coughing, laughing, sneezing or by ***droplets*** airborne diseases *can be transmitted* from infected person to an infected persons.

Continued.....

- Air-borne diseases, obviously, will spread more easily when there is *over crowding* as in overcrowded class rooms, public transport, lounge, dance halls and cinemas.
- *Good ventilation* can do much to counteract the effects of overcrowding.

Air borne disease

- ❖ Definition, agent, incubation period, period of communicability, Epidemiology, risk factor/causes, life cycle, c/m, diagnosis, DDx, management, complications and prevention and control methods
- Measles (rubella)
- Influenza
- Pertussis (whooping cough)
- Diphtheria
- Leprosy (hansen's disease)
- Tuberculosis

Measles

- ❖ **Definition**:-An acute highly communicable viral disease caused by **Measles virus**
- ❖ **Reservoir** – humans (the only natural hosts)
- ❖ ***Epidemiology and occurrence***
 - The disease is very infectious.
 - In general it is a disease *affecting children*.
 - The fatality rate of measles is caused by malnutrition
 - For children in good nutritional condition, measles is a mild disease

Measles...

❖ *Mode of transmission*

- Air borne by droplet spread
- Direct contact with
 - Nasal or throat secretions of infected persons and
 - Less commonly by articles freshly soiled with nose and throat secretions.
- *Greater than 94% of herd immunity may be needed to interrupt community transmission of the disease*

Measles...

❖ ***Incubation period***

- About 10 days, but may be 7-18 days from exposure to onset of fever,
- usually 14 days until rash appears.

a) Complications may result from viral replication or bacterial super infection, and include: otitis media, pneumonia, laryngo tracheo bronchitis (croup), diarrhea and encephalitis.

Measles (clinical manifestation)

- Prodromal fever, conjunctivitis, Coryza, Cough
Leucopenia
- Small spots with white or bluish white centers on an erythematous base on the buccal mucosa (*Koplik spots*).
- A characteristic red blotchy *Rash* appears on the 3rd - 7th day w/h *begins* on face, then becomes generalized, lasts 4-7 days.





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Period of communicability

❖ From 1 day before the beginning of the Prodromal period (usually about ***4 days before rash onset***) to ***4 days after appearance of the rash***; minimal after the second day of rash.

❖ ***Susceptibility and resistance***

❖ All those who are ***non-vaccinated*** or ***have not had the disease*** are susceptible.

❖ ***Permanent immunity*** is acquired ***after natural infection or immunization***.

❖ **Diagnosis**

- ❖ Clinical grounds
- ❖ Koplik spots – typical sign of measles
- ❖ Bright red rash: face → trunk → extremities.

❖ **Differential diagnosis**

- ❖ Upper respiratory tract infections,
- ❖ drug allergic reaction,
- ❖ Scarlet(bright red) fever,
- ❖ chicken pox.

❖ Treatment

- No specific treatment
- Isolate patient
- Treatment of complications
- Aspirin or sponge bath
- Provision of vitamin A
 - 50,000 iu for < 6 months, 100, 000 iu for 6-12 months, 200,000 iu for greater than 12 months. All single dose
- Provision of non-irritant small frequent diet.

Prevention and control

- ❖ Immunization of all children (<5 yrs of age) who had contact with infected children.
- ❖ Provision of measles vaccine at nine months of age.

Influenza

- ❖ **Definition** – An acute viral disease of URT.
- ❖ **Etiology**:-*3 types* of influenza virus (A, B & C)
- ❖ **Reservoir** – *Humans* are the primary reservoirs for human infections.
- ❖ **MOT**– *Air borne spread* predominates among crowded populations in enclosed spaces, such as school, buses; transmission may also occur by direct contact.

➤ **Clinical manifestations**

- ❖ Sudden onset of fever, headache, myalgia, prostration(exhaustion), sore throat and cough.
- ❖ Cough is often severe and protracted but other manifestations are self-limited with recovery in 2-7 days.
- **Period of communicability** – probably 3-5 days from clinical onset in adults; up to 7 days in young children.

➤ Treatment

- i. Antipain and anti pyretic
- ii. High fluid intake, Bed rest
- iii. Balanced diet

➤ ***Prevention and Control***

- i. Educate* the public in basic personal hygiene
- ii. Immunization* with available killed virus vaccines may provide 70-80% protection.

DIPHTHERIA

- **Definition:-** An acute bacterial disease that primarily involves the tonsils, pharynx, larynx, nose, occasionally other mucous membranes of skin and sometimes the conjunctiva or vagina.
- **Etiology** – corynebacterium diphtheria

- **Reservoir** – Humans
- **Mode of transmission** – skin contact with a patient or carrier; more rarely, contact with articles soiled with discharges from lesions of infected people
- ***Incubation period*** – usually 2-5 days

❖ ***C/Manifestation of diphtheria***

- Lesion on the affected area which is asymmetrical adherent grayish white membrane with surrounding inflammation.
- The throat is moderately to severely sore in pharyngo tonsillar diphtheria with cervical lymphnodes.
- In moderate to severe cases, there is marked swelling and edema of the neck with extensive tracheal membranes that progress to air way obstruction.

C/Manifestation of diphtheria...

- Nasal diphtheria can be mild and chronic with one sided nasal discharge and excoriations.
- The toxin can cause Myocarditis, renal failure and neuropathies (cranial & peripheral motor & sensory nerve palsies)

➤ **Period of communicability –**

- variable, until virulent bacilli have disappeared from discharges and lesions; usually ≤ 2 weeks & seldom >4 weeks.

➤ ***Diagnosis***

- i. Based on clinical and epidemiological grounds
- ii. Bacteriologic examination of discharges from lesions

Treatment

- i. Isolation
- ii. Antibiotics
 - a. *Erythromycin* for 2 weeks but 1 week for cutaneous form or
 - b. *Procaine penicillin* for 14 days or
 - c. single dose of *benzathine penicillin*.
- iii. Diphtheria antitoxin

➤ Prevention of diphtheria

- I. *Health education* to the public and particularly the parents of young children, of the hazards of diphtheria and the necessity for active immunization
- II. *Immunization* of infants with diphtheria toxoid.
- III. Concurrent and terminal *disinfection of articles* in contact with patient and soiled by discharges of pt.
- IV. Single dose of *penicillin* (IM) or 7-10 days course of *erythromycin* (PO) is recommended for all persons exposed to diphtheria.

Pertusis (Whooping cough)

- **Definition** – An acute *bacterial* disease involving the respiratory tract.
- **Etiology** – *Bordetella pertusis*, the pertussis bacillus.
- **Epidemiology and occurrence:-** Endemic disease *common to children* especially young children every where in the world.
- ❖ Endemic in developing world & 90% of attacks occur in children < 6 years of age.

- **Reservoir** – *Humans*
- **Mode of transmission** – primarily by *direct contact* with discharges from respiratory mucus membranes of infected persons by air-borne route, probably by droplets.
- *Indirectly* by handling objects freshly soiled with nasopharyngeal secretions.
- **Incubation period** – commonly *7-20 days*

➤ *Clinical manifestations*

- The disease has insidious onset & 3 phases:

A. Catarrhal phase (excessive discharge from nose & throat)

i. Lasts *1-2 weeks*

ii. Insidious onset of *irritative cough* (esp. at night), Coryza, Lacrimation & hacking

B. Paroxysmal phase (sudden onset)

- i. Lasts for *1-2 months* or longer
- ii. Paroxysms are characterized by *repeated violent coughs*.
- iii. ***Each*** series of ***paroxysms has many coughs*** without intervening inhalation and can be followed by a characteristic crowing or high pitched inspiratory whoop.
- iv. Paroxysms frequently end with the expulsion of clear, tenacious mucus, often followed by vomiting.
- v. Cyanosis and sub conjunctival hemorrhage due to violent cough.

- ***C. Convalescent phase***
- The *cough may diminish slowly or may last long.*
- After improvement the disease may recur.

➤ Diagnosis

- i. *C/F* especially in paroxysmal phase
- ii. Isolation of the organism from nasopharyngeal *specimen*.
- iii. *WBC* count >30,000 with marked lymphocytosis

Treatment

❖ Hospitalization of the cases & fluid replacement

- i. Erythromycin* shortens the period of communicability, but does not reduce symptoms except when given during the incubation period.
- ii. Cough suppressants* are only indicated when the child gets too weak to drink and eat and cannot sleep.

iii. Antibiotics for super infections like pneumonia, otitis media.

iv. Suction of mucus from respiratory tract

v. Tracheostomy & tracheal intubation in air way obstruction

LEPROSY (Hansen's disease)



LEPROSY....

- A chronic infectious disease (bacterial) caused by *Mycobacterium leprae*
- The areas most commonly affected are the:-
 - ✓ Superficial peripheral nerves,
 - ✓ Skin,
 - ✓ Mucous membranes of the upper respiratory tract,
 - ✓ Anterior chamber of the eyes

LEPROSY...

❖ Epidemiology

- ✓ Common in tropics and subtropics,
- ✓ Nearly 10 million people are known to be affected worldwide
- ✓ In Ethiopia 5,000 cases are infected each year → highlands of Shoa, Gojjam and Arusi.
- ✓ Prevalence decreasing after MDT
- ✓ Uncommon in infancy
- ✓ Incidence –peak between 10-20 yrs
- ✓ Prevalence-peak 30-50 yrs
- ✓ Males > females (2:1)

LEPROSY....

❖ **Reservoir-** Humans

❖ **Mode of transmission-** Not clearly established.

- Household and prolonged close contact appear to be important.
- Millions of bacilli are liberated daily in the nasal discharges of untreated lepromatous patients.
- Cutaneous ulcers in lepromatous patients may shed large number of bacilli.
- Organisms probably gain access (entrance) through the URT and possibly through broken skin.
- In children less than one year of age, transmission is presumed to be transplacental.

LEPROSY....

- ❖ **Incubation period-** 9 months to 20 years.
- ❖ **Period of communicability-** Infectiousness is lost in most instances within 3 months of continuous and regular treatment with dapsone or clofazamin or within 3 days of rifampicin treatment.

LEPROSY....

- ❖ **Susceptibility and resistance**- The presence and form of leprosy depend on the ability to develop effective cell mediated immunity.
 - majority develop complete resistance and do not proceed to clinical disease
 - If bacilli survive and multiply ,the patient may go on to develop one of the ‘determined’ forms of leprosy .
- ❖ **Clinical Manifestation**
 - Clinical manifestations vary between two polar forms:
 - **lepromatous** and **tuberculoid** leprosy.

LEPROSY....

❖ **Lepromatous (Multibacillary form)**

- Nodules, papules, macules and diffused infiltration are bilaterally symmetrical and usually numerous and extensive.
- Involvement of the nasal mucosa may lead to crusting, obstructed breathing and epistaxis.
- Occular involvement leads to iritis and keratitis.

LEPROSY....

❖ **Tuberculoid (Paucibacillary form)**

- Skin lesions are single or few, sharply demarcated, anesthetic or hyperesthetic and bilaterally symmetrical.
- Peripheral nerve involvement tends to be severe.

LEPROSY....

❖ **Borderline**

- Has features of both polar forms and is more liable to shift toward the lepromatous form in untreated patients and toward the tuberculoid form in treated patients.

LEPROSY....

❖ **Diagnosis**

- Complete skin examination (hyperesthesia, anesthesia, paralysis, muscle wasting or trophic ulcer which are signs of peripheral nerve involvement)
- Skin lesion are tested for sensation (light touch, pink prick, temperature discrimination).
- Demonstration of AFB in skin smears made by scraped incision method.
- Skin biopsy confined to the affected area should be sent for leprosy diagnosis.

Testing feeling in skin patch with cottonwisp



Testing the feeling in the skin patches



Treatment of Leprosy

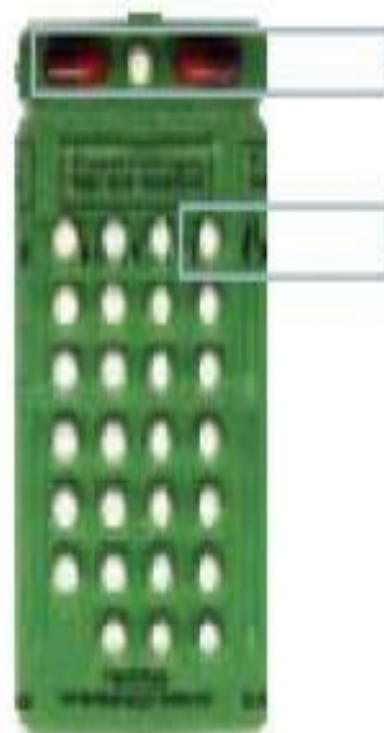
❖ Principles

- ✓ Stop the infection with chemotherapy
- ✓ Prevent and treat reactions and to reduce the risk of nerve damage
- ✓ Educate the patient to cope with existing nerve damage, in particular anesthesia
- ✓ Treat the complications of nerve damage
- ✓ Rehabilitate the patient socially and psychologically.

Chemotherapy for leprosy

- PB Leprosy: (Total duration is 6 months)
 - Dapsone 100mg daily unsupervised
 - Rifampicin 600mg monthly supervised
- MB Leprosy: (Total duration is 12 months)
 - Dapsone 100mg daily unsupervised
 - Clofazimine 50mg daily unsupervised
 - Rifampicin 600mg monthly supervised
 - Clofazimine 300mg monthly supervised

MDT blister packs for adults



PB adult treatment:

Once a month: Day 1

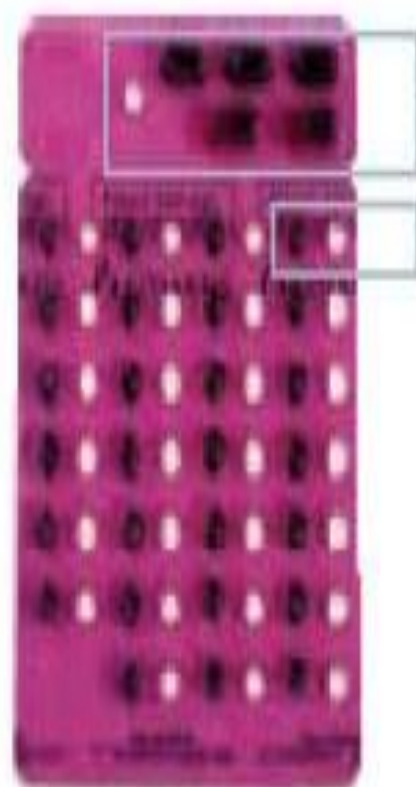
- 2 capsules of rifampicin (300 mg X 2)
- 1 tablet of dapsons (100 mg)

Once a day: Days 2-28

- 1 tablet of dapsons (100 mg)

Full course: 6 blister packs

PB adult blister pack



MB adult blister pack

MB adult treatment:

Once a month: Day 1

- 2 capsules of rifampicin (300 mg X 2)
- 3 capsules of clofazimine (100mg X 3)
- 1 tablet of dapsona (100 mg)

Once a day: Days 2-28

- 1 capsule of clofazimine (50 mg)
- 1 tablet of dapsona (100 mg)

Full course: 12 blister packs

Prevention and control of leprosy

- ✓ Prevent nerve damage
 - Most nerve damage is caused by reactions
 - Early diagnosis, treatment and accurate management of reactions prevent all disabilities
- ✓ Prevent deformity
 - Proper care of skin
 - Proper care of minor wounds & skin infections
 - Provision of shoes
 - Physiotherapy
 - Health education

Prevention and control of leprosy ...

- ✓ Prevent blindness
 - Check for proper eye closure
 - Test the sensation of the cornea
 - Refer patients with red eyes immediately
- ✓ Prevent defaulting Reasons why the patient gives up his treatment
 - No confidence in treatment
 - Clinic too far
 - Shame
 - Disability
 - Thinks he is cured
 - Thinks he has not got leprosy
 - Treatment too long